

THOMAS FARINELLI

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[GitHub](#) | [Portfolio](#)

PROFESSIONAL PROFILE

Master's degree candidate in Computer Engineering, specializing in Artificial Intelligence and Machine Learning, with experience in ML/DL, biomedical data analysis, Python pipelines, statistical analysis, eye tracking, medical imaging and full-stack development. Interested in roles in ML/AI, data science and software engineering.

CORE SKILLS

Programming and data: Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Jupyter, SQL, Git/GitHub

AI / Machine Learning: Machine Learning, Deep Learning, CNNs, transfer learning, model fine-tuning, supervised/unsupervised learning

Biomedical and analytics: Eye-tracking and oculomotor analysis, medical-image analysis, descriptive/inferential statistics, correlation analysis, t-SNE, Grad-CAM

Software development: Java, Spring Boot, SQL, PostgreSQL

Software development - entry level: Spring Cloud Gateway, Spring Cloud LoadBalancer, Consul, Apache Kafka, Docker, Docker Compose, Kubernetes, Gradle, Angular, TypeScript, JavaScript, HTML

PROFESSIONAL EXPERIENCE

Telecommunications Support Technician - Roma Tre University

05/2024 - 07/2026

Rome, Italy

- Managed and resolved technical issues related to audio-video and IT equipment used during teaching activities; checked computers, microphones, projectors and related systems.
- Developed practical skills in troubleshooting, ticket handling, operational prioritization, user support and communication with lecturers and students.

Junior Full Stack Developer - Catenate ICH

11/2023 - 04/2024

Rome, Italy

- Front-end development with Angular, TypeScript and JavaScript, and back-end development with Spring Boot.

Machine Learning Intern - Roma Tre University

06/2023 - 10/2023

Rome, Italy

- Studied machine learning methods for the diagnosis of lung diseases, including Naive Bayes, SVM, Decision Tree, XGBoost, KNN and logistic regression.
- Used evaluation metrics and methods including accuracy, precision, recall, ROC/DET curves and AUC.

SELECTED TECHNICAL PROJECTS

- Oculomotor movement analysis - research code to process eye-tracking recordings, extract fixations, saccades and regressions, and support statistical comparison between presbyopic participants and a normative group.
- Histopathological image classification - group project on preprocessing lung and colon tissue images with ResNet-18, including Wavelet, Retinex/Sobel, CLAHE, transfer learning and Grad-CAM.
- Small Language Model fine-tuning - fine-tuning Phi-3, Mistral and Qwen for structured information extraction using LoRA/QLoRA and schema-based evaluation.

EDUCATION

Master's Degree in Computer Engineering - Artificial Intelligence and Machine Learning

2024 - 2026

- Roma Tre University

Expected graduation: October 2026. Weighted average: 28.19/30; degree-grade average according to the programme rules: 106.42/110 (≈29.02/30). 94/120 ECTS credits completed.

Master's thesis: Analysis of the correlation between visual disorders and data collected through eye tracking.

Bachelor's Degree in Computer Engineering - Roma Tre University

2019 - 2024

Final grade: 103/110. Thesis: Machine learning techniques for the diagnosis of lung diseases.

INTERNATIONAL EXPERIENCE

London, United Kingdom (2017-2019). Life and work experience in an international environment, strengthening autonomy, adaptability, communication skills and collaboration with people from different cultural and professional backgrounds. Available for relocation and international opportunities.

LANGUAGES

Italian: Native | English: B2